

INVERTER FOR SOLDERING IRON

■ LOVELY T.P.

Here is a simple but inexpensive inverter for using a small soldering iron (25W, 35W, etc) in the absence of mains supply. It uses eight transistors and a few resistors and capacitors.

Transistors T1 and T2 (each BC547) form an astable multivibrator that produces 50Hz signal. The complementary outputs from the collectors of transistors T1 and T2 are fed to pnp Darling- ton driver stages formed by transistor

pairs T3-T5 and T4-T6 (utilising BC558 and BD140). The outputs from the drivers are fed to transistors T7 and T8 (each 2N3055) connected for push-pull operation. Use suitable heat-sinks for transistors T5 through T8.

A 230V AC primary to 12V-0-12V, 4.5A secondary transformer (X1) is used. The centre-tapped terminal of the secondary of the transformer is connected to the battery (12V, 7Ah), while the other two terminals of the secondary are connected to the collectors of power transistors T7 and T8,

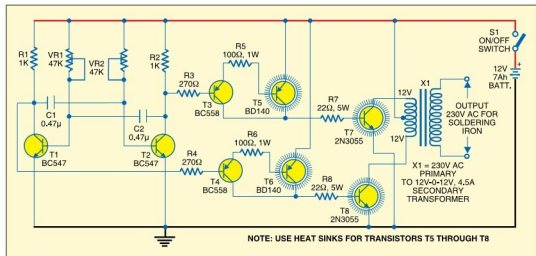


respectively.

When you power the circuit using switch S1, transformer X1 produces 230V AC at its primary terminal. This voltage can be used to heat your soldering iron.

Assemble the circuit on a general-purpose PCB and house in a suitable cabinet. Connect the battery and transformer with suitable current-carrying wires. On the front panel of the box, fit power switch S1 and a 3-pin socket for connecting the soldering iron.

Note that the ratings of the battery, transistors T7 and T8, and transformer may vary as these all depend on the load (soldering iron). ●



RADIATION SENSOR

■ D. MOHAN KUMAR

When you work on a computer or watch TV, your body is engulfed in an 'electronic smog' emanating from the device. For instance, in CRT-based monitors, the spot of electrons that sweep the screen generates pulsed electromagnetic radiation (PEMR). Some of this energy escapes in the form of radiations in very low-frequency and extremely low-frequency energy.

These unnatural radiations can disrupt the body's biomagnetic field within which millions of electrical impulses regulate the activities of the cells. Continuous exposure to PEMR can alter the hormonally controlled biological cycles and cause altered mood, disorientation, eyestrain, stress, etc—generally known as the 'computer vision syndrome'. Sitting close to the computer increases the risk and it is

therefore necessary to maintain safe viewing distance from the monitor.

This circuit helps you to assess the range of radiation from the monitor through LED indicators. The pulsation of radiations can be visualised through the flashing of the LEDs.

The circuit uses a pickup coil (L1) to collect the energy from the radiation. It is a miniature coil transformer used in the compact fluorescent lamp (CFL) circuit. The ends of the coil are connected to the inputs of an operational amplifier (IC TLO71) wired as a differential amplifier.

TLO71 (IC1) is a low-noise JFET

